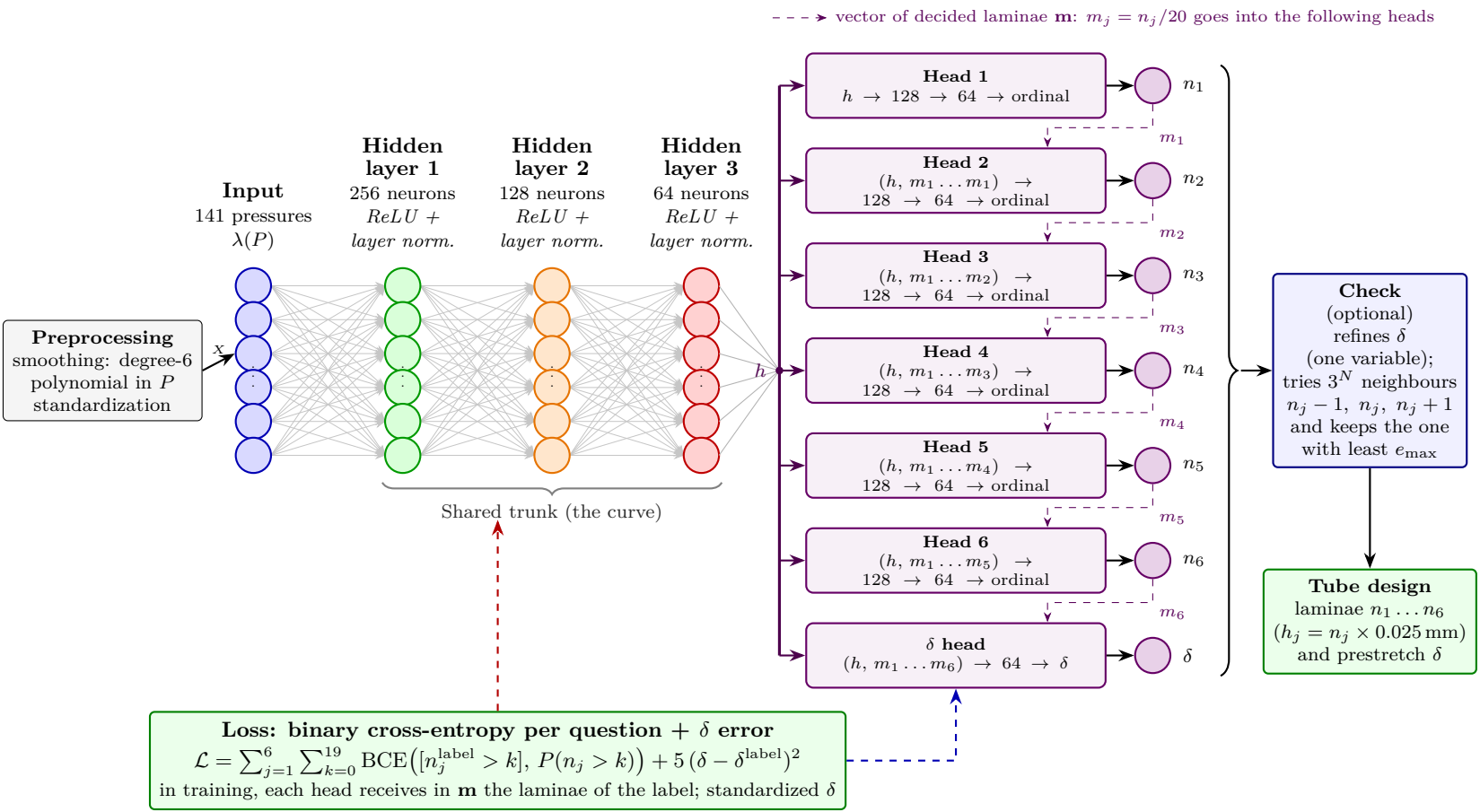
Ordinal chained network for the tube: one head per material and one for δ

trunk $141 \to 256 \to 128 \to 64$; head j: $(h,\, m_1 \dots m_{j-1}) \to 128 \to 64 \to s_j$; 20 questions $P(n_j > k) = \sigma(s_j + b_{jk})$; δ head: $(h,\, \mathbf{m}) \to 64 \to \delta$



Each head answers "more than k laminae?" for $k = 0 \dots 19$; n_j (0 to 20) is the number of "yes". It chooses knowing what the previous heads chose, so it can compensate between materials. It is used in order: material 1, then 2 knowing 1, and so on up to 6; finally, the δ head gives the prestretch knowing all the laminae. About 185 000 parameters (6 materials).